SQL codes

1. Database commands

Create db : create database mydb;

Show all db : show databases;

Use a db : use mydb

Delete a db : drop database mydb;

2. Table commands:

Create table : create table users (

```
    id int auto_increment primary key,
    name varchar (99),
    email varchar (99),
    password varchar (99),
    created_at timestamp default current_timestamp
);
```

show tables : show tables;

View table structure : describe users;

Delete table : drop table users;

3. Insert data :

Inserting multiple rows : insert into users (name, email, password) values ('A', 'abc', '111'), ('B', 'def', '222');

timestamp

4. Read data (select /display)

Select all : select * from users.

Select specific columns : select name, email from users;

Where condition: select * from users where id = 1;

select * from users where name = 'John';

Multiple conditions: select * from users where id > 3 and name = 'A';

select * from users where id < 5 or name = 'B';

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5. Update data:

update users set name = 'Mike', email = 'mike123' where id = 1;

---

6. Delete data :

delete from users where id = 1;

Delete all rows : delete from users ;

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7. Limit , order by

Limit rows: select * from users limit 5;

Order ascending : Select * from users order by name asc;

Order by descending : select * from users order by id desc;

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8. Like (search)

Select * from users where name like 'A %';

" '%n';

" '%oh%';

'222');

9. Aggregate functions:

Count : select count (*) from users;

Max: select max (id) from users;

Min: select min (id) from users;

Sum: select sum (id) from users;

Avg: select avg (id) from users;

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10. Group by + having:

Select name, count (*) from users group by name;

Select name, count (*) from users group by name having count (*) > 1;

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11. Joins:

Inner join: Select users.name, orders.amount from users inner join orders on users.id = orders.user_id;

Left join: Select users.name, orders.amount from users left join orders on user.id = orders.user_id;

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12. Foreign key:

create table orders (id int auto_increment primary key, amount int, user_id int, foreign key (user_id) references users (id));

13. Index :

```
create index idx_email on users(email);
```

14. Constraints :

```
email varchar(100) unique;
name varchar(100) not null.
```

15. Transactions :

```
start transaction;
insert into users values (...);
commit
```

Rollback : Rollback ;

16. User management :

```
create user 'dk'@'localhost' identified by 'psw';
grant all privileges on mydb.* to 'dk'@
                                  'localhost';
flush privileges;
```

17. Backup & restore :

</> Bash - Backup : mysqldump -u root -p
mydb > backup.sql

Restore : mysql -u root -p mydb < backup.sql.